LegalSpan LoRA summarizer: Methodology and Evidence

CS781 Capstone – Krishna K Karthikeya G N

12 November 2026

Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

Keep the legal span representation from Phase 1 and adapt only compact low-rank modules to summarization. The report records selected epoch 4 by negative summary NLL, with no reference-dependent test passage selection.

Architecture and mechanism

The LegalSpan source and tokenizer feed a document-only passage strategy with first/last coverage. LoRA modules modify selected projections, and autoregressive generation produces the summary.

Measured result

Scored submission	1031076, 1031077, 1031078
Codabench BERTScore F1	0.7847
Archived local metric	-1.1865 validation negative NLL; BERTScore not measured in this report

Codabench BERTScore F1 was 0.7847, the highest among these submitted Phase 6 variants. The evidence supports this result for the scored submission, not a universal model ranking.

Limitations and next check

BERTScore is a semantic-overlap measure; legal fact fidelity and citation accuracy require separate review. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: legalspan_lora.py, phase6_data.py, train_phase6.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.